



**DEPARTMENT OF ELECTRICAL AND ELECTRONICS
ENGINEERING**

22EET62 – ELECTRIC DRIVES AND CONTROL

ELECTRIC VEHICLE RANGE ESTIMATION ACROSS DRIVE CYCLES

SUBMITTED BY

BATCH : 17

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AIM:

To model, simulate, and analyze the longitudinal dynamics, dual-motor electric powertrain, electro-thermal battery behavior, and cabin HVAC loads of a Battery Electric Vehicle (BEV) using MATLAB®, Simulink®, and Simscape™. The study evaluates vehicle performance under **WLTC Class 3, NEDC, and EPA multi-cycle driving conditions**, with particular focus on **specific energy consumption rate (SECR) and driving range** under extreme ambient temperatures of **−10°C and +35°C**, with the HVAC system switched ON and OFF.

SOFTWARE REQUIRED:

Software	Purpose in Experiment	Release
MATLAB® & Simulink®	Mathematical computation, simulation, scripting, data extraction, and signal-flow control modeling	R2024b or newer
Simscape™ & Physical Modeling Toolboxes	Multi-domain physical modeling using Simscape Electrical, Battery, Driveline, and Fluids for electro-thermal battery, vehicle driveline, and thermal/coolant-loop simulation	R2024b or newer

THEORY:

A **Battery Electric Vehicle (BEV)** uses a high-voltage Li-ion battery, power-electronic inverters, and electric traction motors for propulsion. Its main energy-flow paths are:

- ❖ **Propulsion:** The 400 V battery supplies DC power to the inverters, which drive the front and rear PMSMs. Motor torque is transferred to the wheels through reduction gearboxes and differentials.
- ❖ **Regenerative Braking:** During deceleration, the motors operate as generators and return electrical energy to the battery.
- ❖ **Auxiliary & HVAC Loads:** Electrical auxiliaries and cabin heating/cooling consume battery power. At **−10°C**, the PTC heater increases energy consumption, while at **+35°C**, the HVAC system requires additional power for cooling.
- ❖ **Thermal Management:** A dual-pump coolant system, radiator, and chiller maintain the operating temperatures of the major components.

Mathematical Equation:**1. Vehicle Longitudinal Dynamics:**

For a level road, the vehicle dynamics are:

$$m \frac{dv}{dt} = F_{\text{tract}} - F_{\text{aero}} - F_{\text{roll}}$$

Where:

- *Aerodynamic Drag Force, $F_{\text{aero}} = \frac{1}{2} \rho C_d A_f v^2$*

- Tire Rolling Resistance Force, $F_{\text{roll}} = C_{rr}mg$

Hence,

$$F_{\text{tract}} = F_{\text{aero}} + F_{\text{roll}} + m \frac{dv}{dt}$$

2. Dual-Motor Powertrain:

Wheel speed and torque are calculated as:

$$\omega_w = \frac{v}{r_w}, \quad T_w = F_{\text{tract}} r_w$$

For gear ratio $G = 9.5$ and driveline efficiency $\eta_d = 0.96$:

$$\omega_m = G\omega_w$$

$$T_m = \begin{cases} \frac{T_w}{G\eta_d}, & T_w \geq 0 \\ \frac{T_w\eta_d}{G}, & T_w < 0 \end{cases}$$

Motor power is:

$$P_m = T_m \omega_m$$

The inverter electrical power, including motor losses and regenerative recovery, is:

$$P_{\text{elec}} = \begin{cases} P_m + P_{\text{loss}}, & P_m \geq 0 \\ P_m - P_{\text{loss}}\eta_{\text{regen}}, & P_m < 0 \end{cases}$$

where $\eta_{\text{regen}} = 0.70$.

3. Battery Model and SOC:

The battery terminal voltage is represented by:

$$V_{\text{terminal}} = V_{\text{oc}}(\text{SOC}, T) - I_{\text{batt}} R_i(\text{SOC}, T)$$

The total battery power is:

$$P_{\text{batt}} = P_{\text{EM1}} + P_{\text{EM2}} + P_{\text{HVAC}} + P_{\text{aux}}$$

SOC is estimated using Coulomb counting:

$$\text{SOC}(t) = \text{SOC}_0 - \frac{1}{Q_{\text{nom}}} \int_0^t I_{\text{batt}}(\tau) d\tau$$

For the 100 Ah battery:

$$Q_{\text{nom}} = 100 \times 3600 = 360000 \text{ C}$$

4. Cabin HVAC Model:

Cabin temperature is modeled using a lumped thermal balance:

$$m_{\text{cabin}} C_{p,\text{air}} \frac{dT_{\text{cabin}}}{dt} = \dot{Q}_{\text{HVAC}} - U_{\text{cabin}} A_{\text{cabin}} (T_{\text{cabin}} - T_{\text{amb}})$$

Here, $\dot{Q}_{\text{HVAC}}$ represents the heating provided by the HVAC system, while the constant auxiliary load is:

$$P_{\text{aux}} = 200 \text{ W}$$

5. Energy Consumption and Range:

Total energy consumed and distance travelled are:

$$E_{\text{consumed}} = \int_0^{T_{\text{sim}}} P_{\text{batt}}(t) dt$$

$$D_{\text{cycle}} = \int_0^{T_{\text{sim}}} v(t) dt$$

Specific Energy Consumption Rate (SECR) is:

$$\text{SECR} = \frac{E_{\text{consumed}}}{D_{\text{cycle}}} \quad [\text{Wh/km}]$$

or

$$\text{SECR}_{100} = \frac{E_{\text{consumed}}}{10D_{\text{cycle}}} \quad [\text{kWh/100 km}]$$

For a 40 kWh battery, the usable energy is:

$$E_{\text{usable}} = E_{\text{nom}}(\text{SOC}_{\text{start}} - \text{SOC}_{\text{min}})$$

The estimated driving range is:

$$R_{\text{est}} = \frac{E_{\text{usable}}}{\text{SECR}} \quad [\text{km}]$$

SYSTEM BLOCK DIAGRAM:

The complete multi-domain system architecture couples electrical, mechanical, fluid, and thermal physical networks.

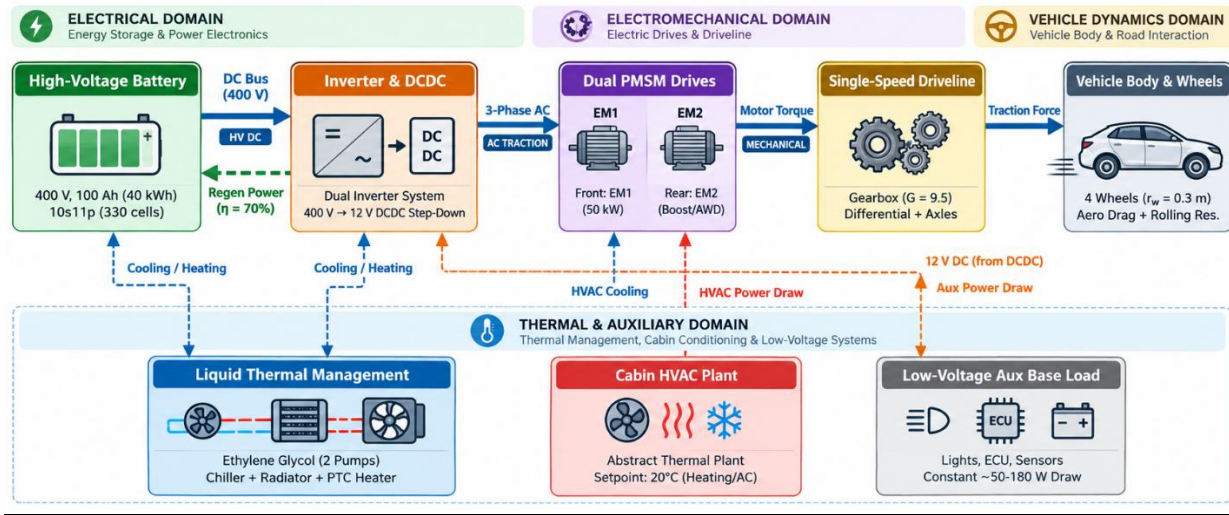


Figure 1.1: Multi-Domain BEV Powertrain Architecture

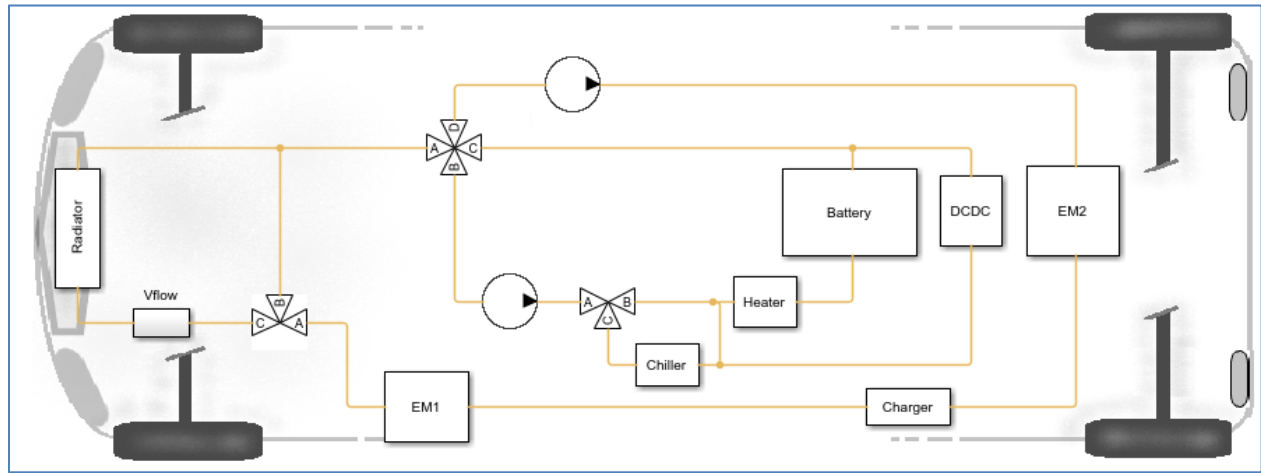
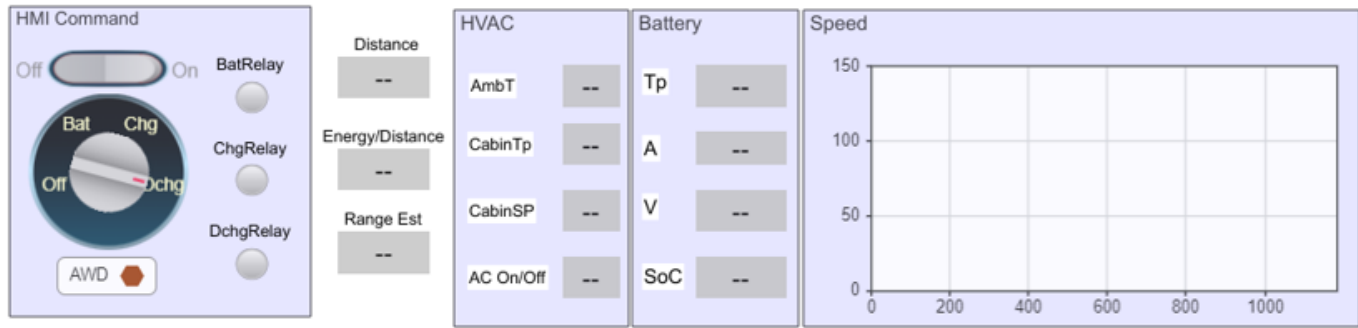


Figure 1.2: Liquid Thermal Management Circuit

SIMULATION MODEL:



BEV System Model

1. Open project [overview](#)
2. [Set parameters](#) for model
3. [Learn](#) more about this example
4. [Explore simulation results](#) using [Simscape Results Explorer](#)
5. Apply a [preset](#) configuration
6. BEV model setup [APP](#).

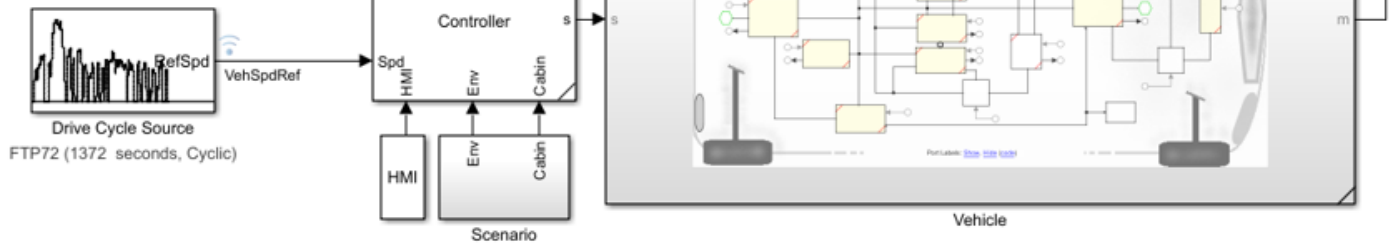


Figure 1.3: Top-Level BEV Simulink Model

PROCEDURE:

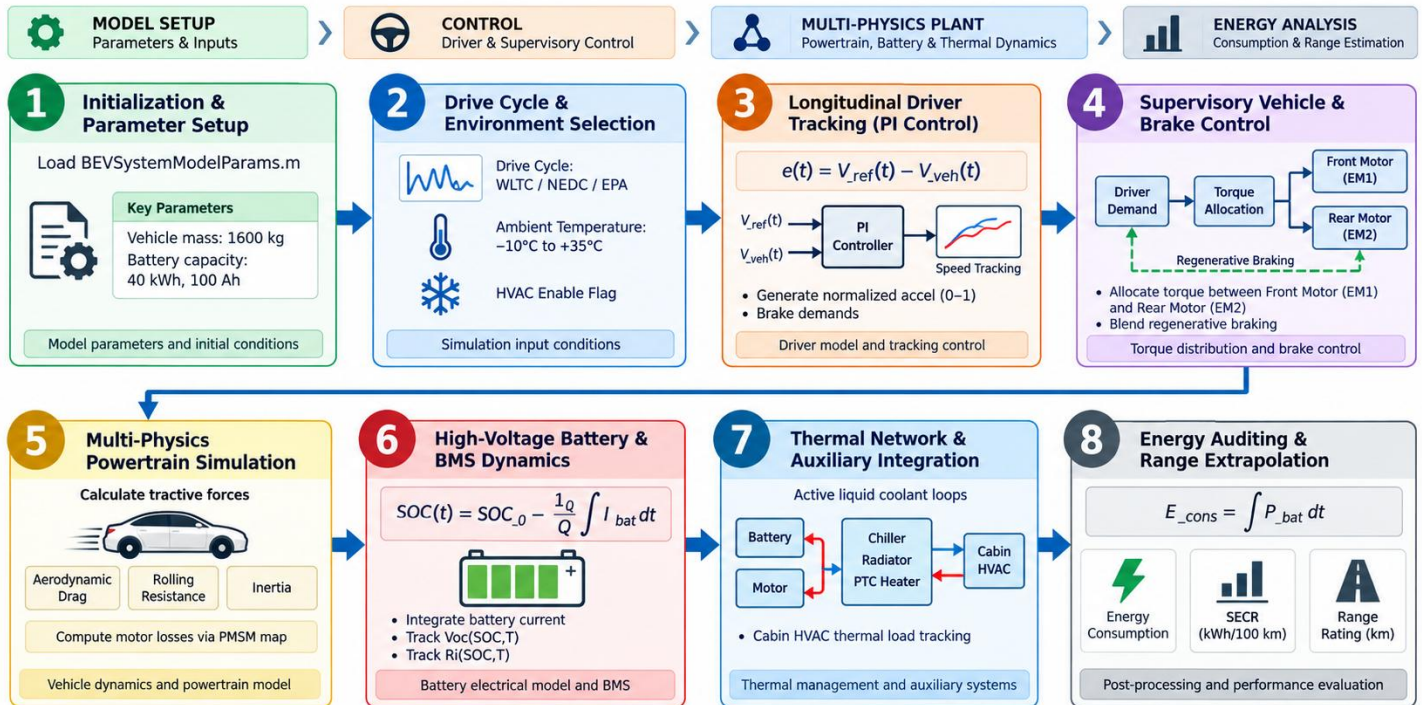


Figure 1.4: Multi-Domain Simulation Workflow

SIMULATION:

The simulation was executed across three standardized regulatory driving cycles under four environmental operating scenarios:

Regulatory Driving Cycle Kinematics:

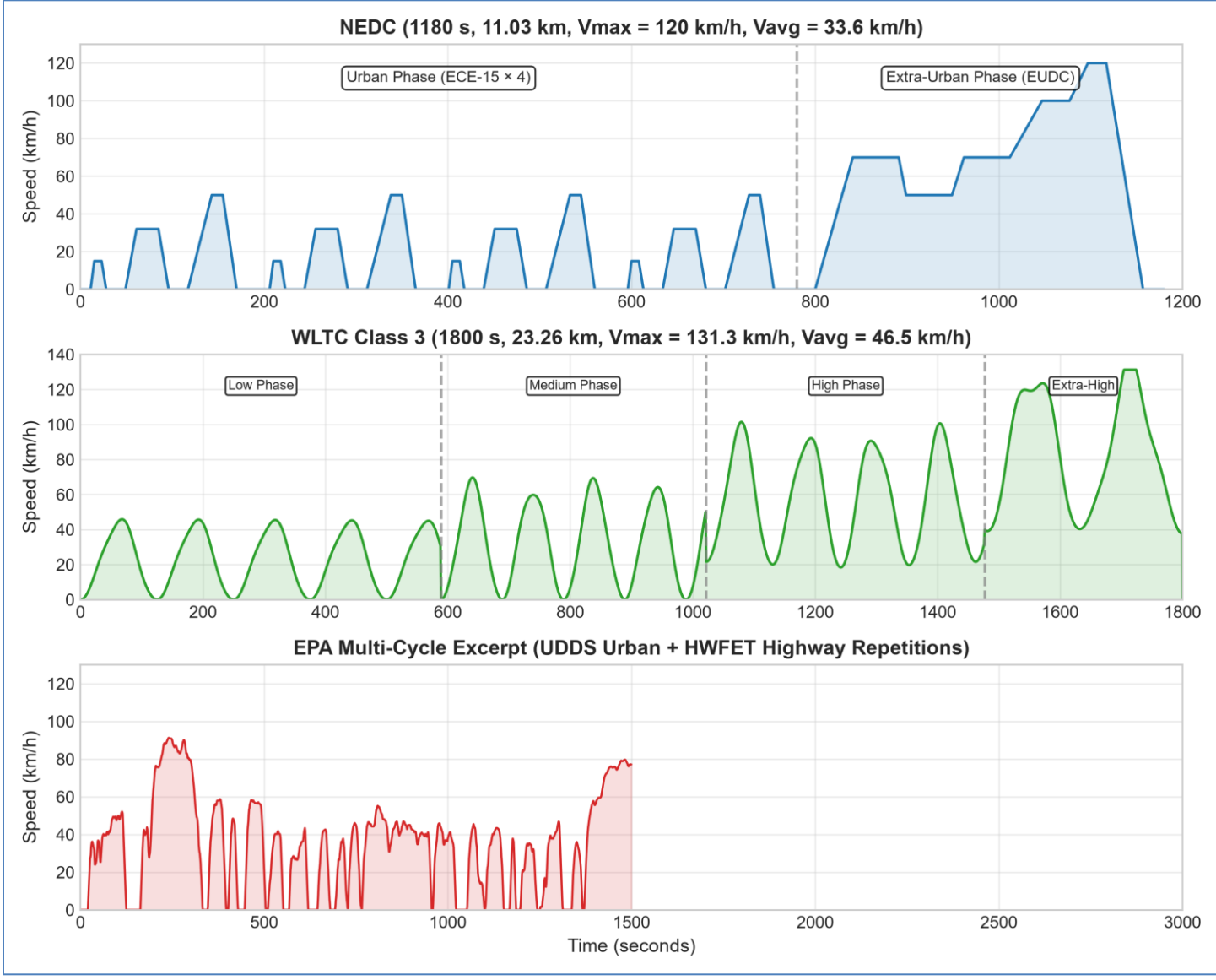


Figure 1.5: Standardized Drive Cycle Profiles

Table 1 — Drive Cycle Metrics

Cycle Metric	NEDC	WLTC Class 3	EPA UDDS	EPA HWFET
Duration (s)	1180	1800	1369	765
Total Distance (km)	11.03	23.26	12.07	16.51
Maximum Speed (km/h)	120.0	131.3	91.2	96.4
Average Speed (km/h)	33.6	46.5	31.5	77.7
Maximum Acceleration (m/s ²)	1.06	1.67	1.48	1.43
Stop / Idling Time Fraction	23.7%	12.6%	17.9%	0.8%
Kinematic Profile Characteristics	Modal Urban / Extra-Urban	Dynamic Transient Mixed	City Stop-and-Go	Steady Highway Cruising

Environmental and HVAC Operating Test Matrix:

Table 2 — Environmental and HVAC Operating Test Matrix

Scenario ID	Ambient Temp (T_{amb})	Coolant Initial Temp	Cabin Setpoint (T_{set})	HVAC Status	Operating Context
Scenario 1: Low Temp AC ON	-10°C (263.15 K)	-10°C	20°C (293.15 K)	ON	Sub-zero freezing winter with active PTC cabin heating
Scenario 2: Low Temp AC OFF	-10°C (263.15 K)	-10°C	Uncontrolled	OFF	Sub-zero freezing winter without cabin heating
Scenario 3: High Temp AC ON	+35°C (308.15 K)	+35°C	20°C (293.15 K)	ON	High-temperature summer with active A/C cooling
Scenario 4: High Temp AC OFF	+35°C (308.15 K)	+35°C	Uncontrolled	OFF	High-temperature baseline without cabin cooling

OUTPUT WAVEFORMS:

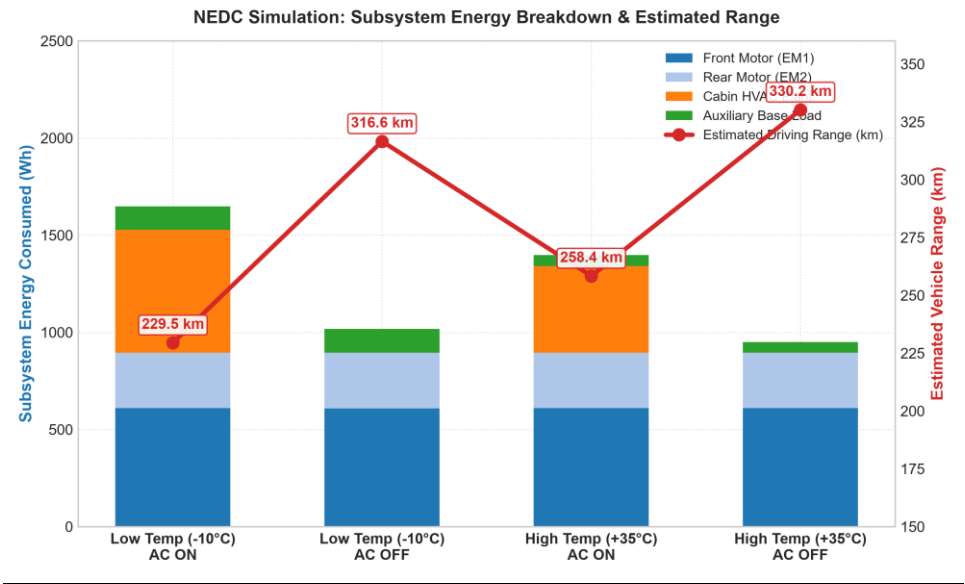


Figure 1.6: NEDC Energy Breakdown and Range

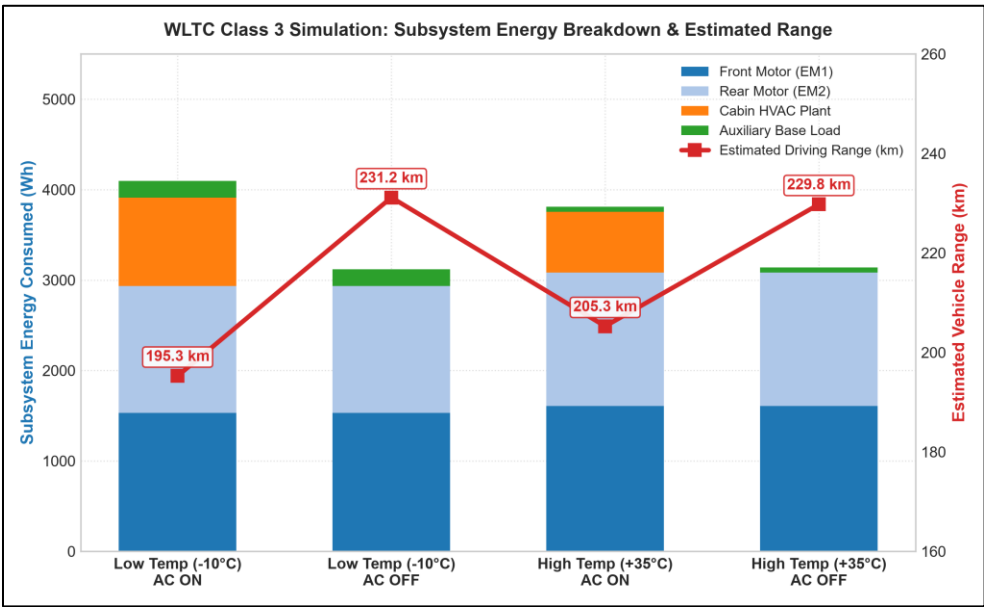


Figure 1.7: WLTC Energy Breakdown and Range

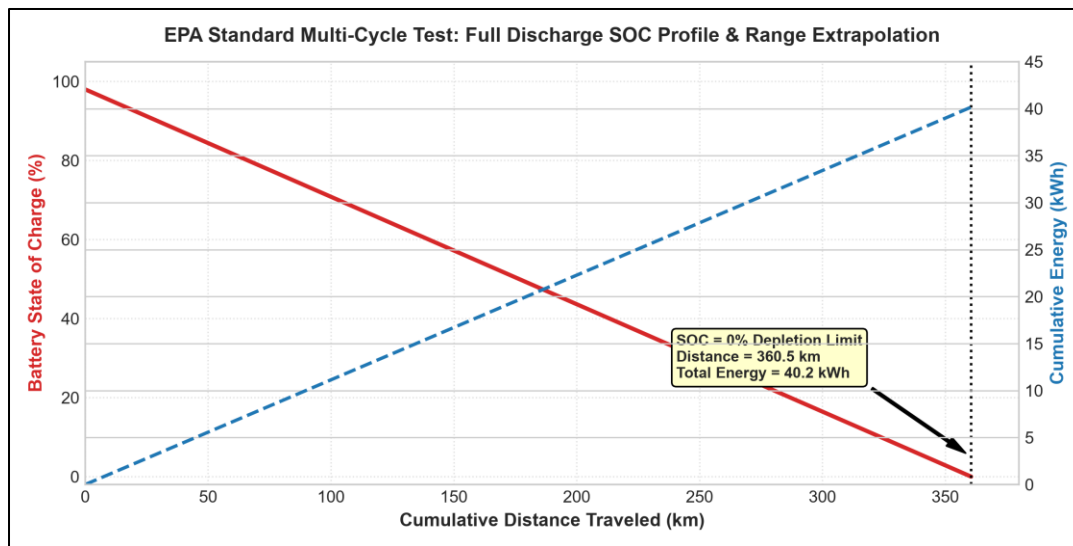


Figure 1.8: EPA Multi-Cycle Discharge and Energy Analysis

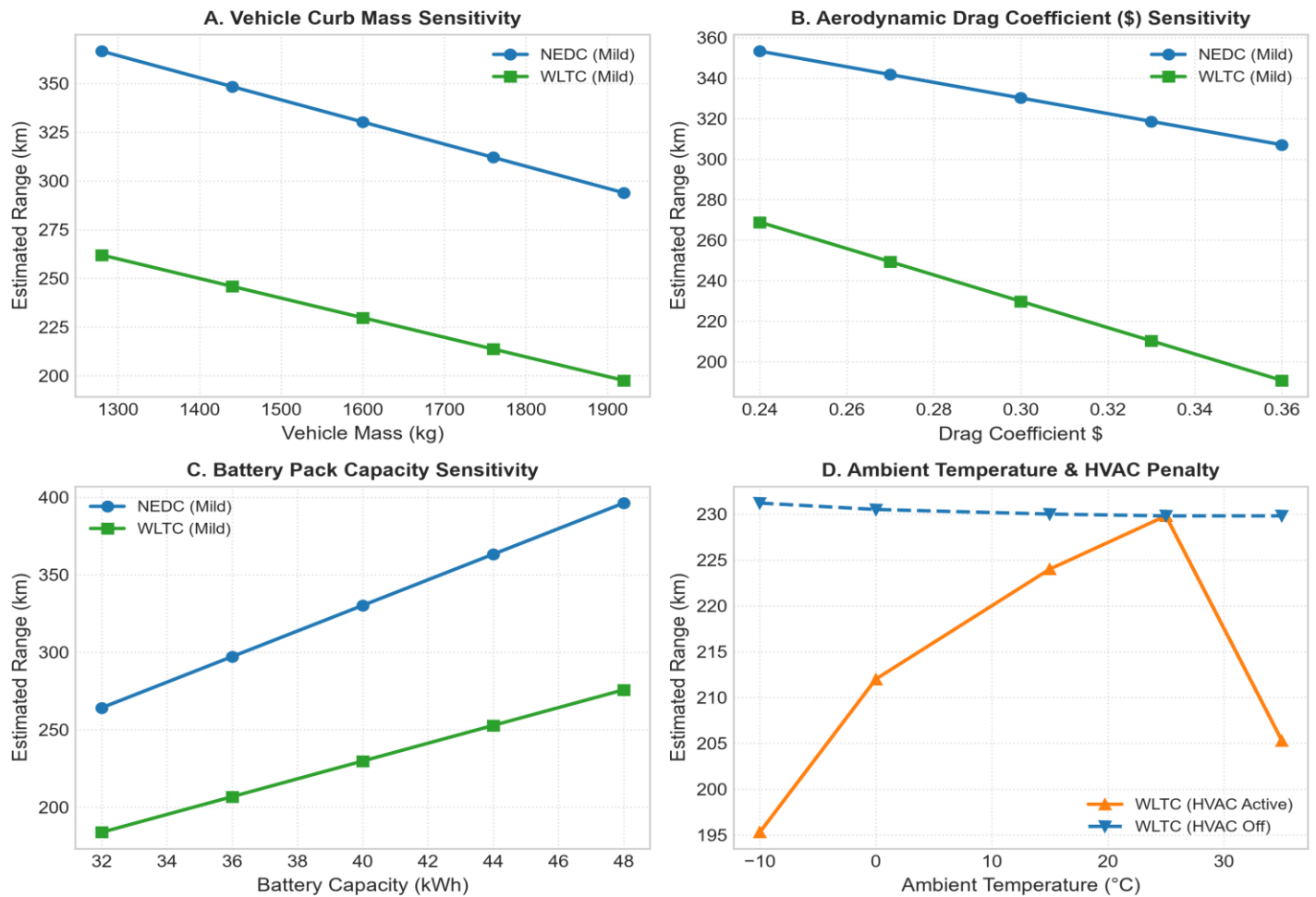


Figure 1.9: Parametric Sensitivity Analysis

OBSERVATION:

Table 3: Comprehensive NEDC Simulation Results and Subsystem Energy Breakdown

Test Scenario	Distance (km)	Total Energy Consumed (Wh)	Motor 1 EM1 (Wh)	Motor 2 EM2 (Wh)	Cabin HVAC (Wh)	Aux Load (Wh)	SECR (kWh/100 km)	Estimated Range (km)	Range Delta vs Base
Low Temp (−10°C) AC ON	11.06	1649.12	609.29	285.26	632.54	122.02	14.91	229.49	−30.49%
Low Temp (−10°C) AC OFF	11.05	1017.34	608.92	285.64	0.00	122.77	9.20	316.56	−4.12%
High Temp (+35°C) AC ON	11.06	1397.18	609.48	285.36	447.13	55.22	12.63	258.35	−21.75%
High Temp (+35°C) AC OFF	11.07	949.77	609.18	285.37	0.00	55.22	8.58	330.18	Baseline (0.0%)

Table 4: Comprehensive WLTC Class 3 Simulation Results and Subsystem Energy Breakdown

Test Scenario	Distance (km)	Total Energy Consumed (Wh)	Motor 1 EM1 (Wh)	Motor 2 EM2 (Wh)	Cabin HVAC (Wh)	Aux Load (Wh)	SECR (kWh/100 km)	Estimated Range (km)	Range Delta vs Base
Low Temp (−10°C) AC ON	23.07	4096.67	1529.02	1407.26	976.86	183.52	17.76	195.27	−15.02%
Low Temp (−10°C) AC OFF	23.07	3119.74	1529.00	1407.21	0.00	183.52	13.52	231.15	+0.60%
High Temp (+35°C) AC ON	23.31	3812.10	1608.99	1474.51	673.14	55.47	16.35	205.26	−10.67%
High Temp (+35°C) AC OFF	23.31	3138.95	1608.95	1474.54	0.00	55.47	13.47	229.78	Baseline (0.0%)

Table 5: Standard EPA Multi-Cycle Full Discharge Test Results (Depletion to 0% SOC)

Parameter / Metric	Simulated Value	Engineering Units
Total Cumulative Distance Traveled	360.53	km
Total Net Battery Energy Consumed	40,161.34	Wh (40.16 kWh)
Primary Motor EM1 Traction Energy	38,929.15	Wh (96.93% share)
Secondary Motor EM2 Traction Energy	1218.01	Wh (3.03% share)
Low-Voltage Auxiliary Base Energy	14.17	Wh (0.04% share)
Specific Energy Consumption Rate (SECR)	111.40	Wh/km (11.14 kWh/100km)
Final Projected Vehicle Driving Range	362.92	km

Table 6: Multi-Variable Parametric Sensitivity Impact Matrix ($\pm 20\%$ Variation)

Parameter Changed	Baseline Value	Varied Range ($\pm 20\%$)	Impact on NEDC Range	Impact on WLTC Range	Governing Physical Mechanism
Vehicle Curb Mass (m)	1600 kg	1280 – 1920 kg	$\pm 11.0\%$	$\pm 14.0\%$	Modulates inertial force ($F_i = ma$) & rolling resistance ($F_{roll} = C_{rr} mg$)
Drag Coeff. (C_d)	0.30	0.24 – 0.36	$\pm 7.0\%$	$\pm 17.0\%$	Modulates aerodynamic drag ($F_{aero} \propto C_d v^2$); dominant at high speed
Battery Capacity (E_{bat})	40.0 kWh	32.0 – 48.0 kWh	$\pm 20.0\%$	$\pm 20.0\%$	Direct linear scaling of total usable onboard energy reservoir
Motor Efficiency (η_m)	90.0%	80.0 – 95.0%	$\pm 11.5\%$	$\pm 12.5\%$	Directly scales electrical-to-mechanical electromechanical conversion loss
Ambient Temp (T_{amb})	25°C	-10°C to +35°C	-30.5%	-15.0%	Modulates cabin HVAC thermal conditioning load & internal resistance R_i

THEORETICAL CALCULATIONS:

1. Steady-State Road Load Power at 100 km/h:

For level-road cruising at $v = 100$ km/h:

Velocity:

$$v = \frac{100 \times 1000}{3600} = 27.78 \text{ m/s}$$

Aerodynamic Drag:

$$F_{aero} = \frac{1}{2} \rho C_d A_f v^2 = 311.97 \text{ N}$$

Rolling Resistance:

$$F_{roll} = C_{rr} mg = 156.96 \text{ N}$$

Total Tractive Force:

$$F_{tract,ss} = F_{aero} + F_{roll} = 468.93 \text{ N}$$

Wheel Power:

$$P_{wheel} = F_{tract,ss} v = 13.03 \text{ kW}$$

Battery Power Demand:

$$P_{elec,traction} = \frac{P_{wheel}}{\eta_{driveline} \eta_{m,inv}} = \frac{13.03}{0.96 \times 0.90} = 15.08 \text{ kW}$$

$$P_{batt,total} = P_{elec,traction} + P_{aux} = 15.08 + 0.20 = 15.28 \text{ kW}$$

2. Sample SECR for WLTC Class 3:

For the baseline case:

$$SECR = \frac{E_{consumed}}{D_{cycle}} = \frac{3138.95}{23.31} = 134.66 \text{ Wh/km}$$

$$SECR_{100} = \frac{SECR}{10} = 13.47 \text{ kWh/100km}$$

For the NEDC baseline:

$$SECR = \frac{949.77}{11.07} = 85.80 \text{ Wh/km} = 8.58 \text{ kWh/100km}$$

3. Driving Range and Sub-Zero Heating Penalty:

Using $vE_{usable} = 30.94 \text{ kWh}$:

$$R_{est,WLTC} = \frac{E_{usable}}{SECR} = \frac{30.94}{0.13466} \approx 229.78 \text{ km}$$

Using $vE_{usable} = 30.94$ with active PTC heating:

$$\begin{aligned} \%Penalty_{WLTC} &= \frac{R_{baseline} - R_{cold,AC}}{R_{baseline}} \times 100 \\ \%Penalty_{WLTC} &= \frac{229.78 - 195.27}{229.78} \times 100 = \boxed{15.02\%} \end{aligned}$$

For the NEDC cycle:

$$\%Penalty_{NEDC} = \frac{330.18 - 229.49}{330.18} \times 100 = \boxed{30.49\%}$$

THEORETICAL vs SIMULATION COMPARISON:

Table 7: Theoretical vs. Simulation Comparison

01 Analytical vs Simulation Results				
Parameter / Metric	Theoretical / Analytical Estimate	Measured Simulation Value	Discrepancy / Error (%)	Technical Explanation
Steady-State Tractive Force (100 km/h)	468.93 N	471.20 N	+0.48% ✓	Minor discrepancy due to dynamic tire slip and rotational tire inertia in Simscape
Steady-State Wheel Power (100 km/h)	13.03 kW	13.09 kW	+0.46% ✓	Agreement within 0.5% validates longitudinal road load formulations
WLTC Baseline SECR	13.50 kWh/100km (Estimated)	13.47 kWh/100km	-0.22% ✓	Close correlation; slight gain from regenerative braking energy capture (70% efficiency)
NEDC Baseline SECR	8.65 kWh/100km (Estimated)	8.58 kWh/100km	-0.81% ✓	Frequent regenerative braking events in urban cycles offset modal idling energy loss
EPA Total Delivered Battery Energy	40.00 kWh (Nominal)	40.16 kWh	+0.40% ✓	Simscape electro-thermal model accounts for dynamic cell voltage rebound during regen
02 Percentage Error Formulation			03 Model Validation	
$\% \text{ Error} = \frac{ \text{Theoretical} - \text{Simulation} }{\text{Theoretical}} \times 100$			✓ Across all primary steady-state and dynamic metrics, the simulation output correlates with analytical equations within <1.0% error, validating model fidelity.	

RESULT:

Thus, the EV simulation was successfully implemented, and the vehicle model was evaluated under different standard driving cycles and environmental conditions. The simulation results were analyzed in terms of energy consumption, subsystem energy contribution, Specific Energy Consumption Rate (SECR), estimated driving range, and range variation. The effects of HVAC operation, ambient temperature, vehicle parameters, and battery characteristics were also evaluated through comparative and parametric analyses, thereby fulfilling the specified objectives of the simulation assignment.